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Abstract:	The valuation of pre-launch Decentralized Autonomous Organization (DAO) tokens is challenging due to both their dual nature—serving as currency and governance rights—and the lack of market data. In this paper, we propose a novel valuation model for pre-launch DAO tokens by employing an agent-based simulation. By leveraging an artificial financial market driven by simulated economic agents, we aim to overcome the difficulty in obtaining a fair valuation for pre-launch DAO tokens. After extending a stochastic differential equation model from prior work, we calibrated our model parameters using Ethereum (ETH) as a case study. Our results suggest that the agent-based simulation method can be a viable computational alternative for valuing pre-launch DAO tokens, aiding investors and DAO developers in selecting suitable tokenomics designs. Furthermore, the proposed agent-based framework is highly flexible and can be easily adapted by modifying agent payoff functions and incorporating DAO-specific features to suit individual DAO needs.

Highlights

- Used agent-based model to value Decentralized Autonomous Organization (DAO) token
- Extend stochastic differential equation model by adding key DAO market features
- Agent-based simulation captures the complex nature of DAO token market
- Model can be calibrated using real-world DAO token market data
- Model provides pre-launch token valuation and suggests optimal tokenomics design

An Agent-based Approach to Valuation of Pre-Launch Decentralized Autonomous Organization Tokens

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Abstract

The valuation of pre-launch Decentralized Autonomous Organization (DAO) tokens is challenging due to both their dual nature—serving as currency and governance rights—and the lack of market data. In this paper, we propose a novel valuation model for pre-launch DAO tokens by employing an agent-based simulation. By leveraging an artificial financial market driven by simulated economic agents, we aim to overcome the difficulty in obtaining a fair valuation for pre-launch DAO tokens. After extending a stochastic differential equation model from prior work, we calibrated our model parameters using Ethereum (ETH) as a case study. Our results suggest that the agent-based simulation method can be a viable computational alternative for valuing pre-launch DAO tokens, aiding investors and DAO developers in selecting suitable tokenomics designs. Furthermore, the proposed agent-based framework is highly flexible and can be easily adapted by modifying agent payoff functions and incorporating DAO-specific features to suit individual DAO needs.

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1. Introduction

Since Nakamoto (2008), blockchain technology has experienced remarkable growth in popularity and adoption across diverse sectors such as finance, supply chain management, and digital identity verification. A notable example is the emergence of the decentralized autonomous organization (DAO), which leverages blockchain and smart contracts to operate autonomously and transparently without a central authority (Buterin, 2014; Wang et al., 2019; El Faqir et al., 2020). In contrast to traditional organizations which rely on a central authority representing shareholders, DAOs are governed collectively through a token-based voting system (Han et al., 2023). This token-based system not only democratizes the governance process but also introduces complex factors that influence DAO token value. The function of DAO tokens as a currency within the

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platform (Cong et al., 2021), providing utility value to holders and simultaneously representing ownership or governance rights, heightens the importance of an endogenously evolving demand–supply balance. This balance is shaped not only by the platform’s growth under market mechanisms but also by the token supply mechanisms determined by the platform’s stakeholders. As a result, changes in token demand and supply dynamics can reverberate through the ecosystem, making it challenging to apply traditional equity valuation methods. Furthermore, as argued in Johnson et al. (2019), obtaining a fair valuation of such tokens is difficult especially in their early stage. To address these challenges, this paper proposes an alternative method for valuing DAO tokens before their launch.

Several studies have explored the valuation of digital tokens within blockchain ecosystems. Cong et al. (2021) proposed a dynamic valuation model defined by a set of stochastic differential equations, leading to a substantial body of work based on this model (Mayer, 2021; Cong et al., 2022a, 2022b; Shen et al., 2023a, 2023b; Sockin & Xiong, 2023, Prat et al., 2024). While Cong et al. (2021) focused on the characteristics of tokens as investable assets and a method of payment within these platforms, Han et al. (2023) introduced a model focusing on the value of tokens as a means of voting. Their model investigated how the distribution of tokens affects the governance value and the price of the token. Other works including Johnson et al. (2019) and Xu et al. (2022) applied traditional valuation methods to cryptocurrencies, while Liu & Zhang (2023) utilized statistical models in cryptocurrency valuation. This paper builds upon existing research by introducing an agent-based valuation model that is designed to handle the complexity observed in the cryptocurrency market.

The application of agent-based models (ABMs) in economics and finance is well-established. Studies by Bullard & Duffy (1998), Arthur (2006), Assenza et al. (2015), and Poledna et al. (2023) have discussed the use of an agent-based framework in understanding or forecasting economic dynamics. Lebaron (1999), Day & Huang (1990), Beja & Goldman (1980), and Palmer et al. (1994) are some early examples of agent-based artificial markets. Lebaron (1999)’s construction of the well-known Santa Fe Artificial Stock Market is especially acknowledged as a pioneering work in this field, and numerous studies of agent-based financial markets followed (Lux & Marchesi, 2000; Raberto et al., 2001; Raberto et al., 2005; Chiarella et al., 2009; Iori & Porter, 2012; Arthur et al., 2018). A detailed review of the artificial financial markets is provided in Tesfatsion & Judd (2006) and Hommes (2006). Although agent-based artificial markets have been shown to reproduce stylized facts in the field of finance (Cont, 2007), many of the recent studies emphasize the cruciality of validation and calibration procedures of the agent-based financial market for more accurately reproducing real-life market characteristics (Fabretti, 2011, 2013; Li et al., 2014; Recchioni et al., 2015; Platt & Gebbie, 2016; Barde, 2016; Lamperti et al., 2018; Lux & Zwinkels, 2018; Platt, 2020; Zhang et al., 2020; Zegadlo, 2021; Dyer et al., 2023). These studies often utilize models proposed by Brock & Hommes (1998), Farmer & Joshi (2002), or Chiarella et al. (2009) and provide valuable resources for calibrating an agent-based financial market model.

Despite this robust foundation of ABM in economics and finance, its application within the cryptocurrency industry remains relatively scarce. Studies such as those by

Hülsemann & Tumasjan (2019), Karra et al. (2023), Fernandez et al. (2024), and Cocco et al. (2017) stand out as some of the few that employ ABMs to analyze the various components of the cryptocurrency industry. Hülsemann & Tumasjan (2019) constructed an ABM to investigate how different token designs affect users' activities. Karra et al. (2023) analyzed mining activities and token supply circulation in a decentralized platform using an ABM. Fernandez et al. (2024) studied the concentration of governance tokens through an agent-based simulation of trading activities among agents. Cocco et al. (2017) proposed an ABM primarily representing the Bitcoin market, and they claimed that the simulated series was aligned with the Bitcoin price dynamics and reproduces stylized facts of the financial market.

Building on existing research, this paper addresses a notable gap in current application of ABMs to the valuation of DAO tokens. Primarily based on the model provided in Cong et al. (2021), we propose an ABM for DAO token valuation. Specifically, we first construct an agent-based representation of the stochastic model proposed by Cong et al. (2021) to introduce market complexity into the valuation model. The connection between stochastic models and ABMs was studied in Feng et al. (2021). However, the Cong et al. (2021) model comes with limitations beyond market complexity as well; it does not incorporate dynamic market size, dynamic token supply, value of tokens as governance right, and token mechanisms such as locking and staking. These components are essential to capture the effect of evolving demand-supply balance and the dual nature of DAO tokens as both an ownership and a currency. To overcome these limitations, we extend the agent-based representation of the Cong et al. (2021) model, incorporating additional components to align better with reality. After developing the agent-based token valuation model, we perform a calibration case study using real-world DAO data, demonstrating the model's ability to generate price forecasts. The case study will mainly focus on DAO token prices immediately after launch and upon their entry into the market. Our focus is on the early stage of DAO launch since the lack of historical price data complicates fair DAO token pricing, which may lead to a sharp price drop in the future. Gaining a deeper understanding of different tokenomics designs on the early DAO token price is crucial for early investors and DAO designers in reducing the uncertainty inherent in pre-launch DAO. By utilizing the capabilities of ABM, we capture the complexities of the market, enhancing our analysis beyond conventional equation-based approaches. We implemented the proposed ABM in Netlogo (Wilensky, 1999).

2. Model

In this section, we introduce our proposed framework for valuing DAO tokens, which addresses the difficulty of applying traditional equity valuation methods to tokens that function both as currency and governance rights. In doing so, we aim to fill the gaps left by existing research—particularly that of Cong et al. (2021), which primarily focuses on payment and investment attributes but does not fully consider the token's dual nature. Our approach starts by adapting the stochastic model of Cong et al. (2021) into an agent-based representation featuring a order book mechanism inspired by Cocco et al. (2017). We

further modify this representation to incorporate dynamic market size, dynamic token supply, and staking and locking functionalities that capture the shifting demand–supply landscape of DAO ecosystems. Moreover, we incorporate governance value and voting participation—elements largely absent in Cong et al. (2021)—to better represent the token’s ownership dimension. By weaving these additional components into the model, we aim to provide a more robust and comprehensive approach to DAO token valuation, particularly for early-stage tokens with limited historical price data.

2.1 A Model for DAO Token with Utility Value

First, we consider a DAO token that offers certain utility values to holders, including access to a product or service offered by the DAO and enabling economic transactions within the platform. Following Cong et al. (2021), the token thus has value as a means of payment and benefits from network effects. We interpret the token’s utility broadly, acknowledging that each DAO token provides a variety of benefits to holders.

2.1.1 The Platform Productivity

Following Cong et al. (2021) which addresses a similar context and objective to our model, we characterize the platforms’ exogenous technological, political, and economic fluctuations using a geometric Brownian motion for the productivity measure A_t :

$$\frac{dA_t}{A_t} = \mu^A dt + \sigma^A dZ_t \quad (1)$$

μ^A and σ^A are parameters of the underlying standard Brownian motion Z_t . The productivity measure in the proposed ABM governed by this equation at each time step. Specifically, for all $t > 0$, the platform productivity measure is updated using the following formula:

$$A_t = A_{t-1} \exp\left(\left(\mu^A - \frac{\sigma^A}{2}\right)dt + \sigma^A Z_t \sqrt{dt}\right) \quad (2)$$

The parameters μ^A , σ^A and an initial level of platform productivity A_0 are controllable parameters, and they must be chosen based on the assumptions underlying the DAO token in question. The estimation results of μ^A , σ^A , and A_0 for real-world DAO tokens are provided in the calibration case study section.

2.1.2 The Market mechanism

The trading mechanism utilized in our model resembles that of Cocco et al. (2017), which modeled a Bitcoin market using an agent-based framework. Raberto et al. (2005) argued that adopting the limit order book mechanism allows the model to reproduce macro-level behaviors. Given its ability to approximate realistic trading mechanisms and the fact that some of the major cryptocurrency exchanges including Binance, Coinbase Exchange,

and Bybit utilize the order book mechanism, we believe that order book mechanism sufficiently approximates the characteristics of a cryptocurrency exchange. Therefore, in the proposed model, each agent places buy and sell orders through an order book, which simulates a realistic token transactions. The orders in the model contain the limit price, and for simplicity they expire after one trading round. Thus, each agent is asked to update their buy and sell orders every trading round based on its perception of the token value. These simplifications serve to speed up the simulation and to avoid introducing unnecessary parameters to the calibration process. Short selling is not allowed in our model due to simplicity, following Cocco et al. (2021).

During each trading round, the order book executes all orders and calculates the latest price of the token. The execution procedure is summarized in Figure 1 and 2. The trading round ends once all matching orders are executed and each agent's token account is updated. After the round concludes, unmatched orders are deleted from the order book, and the model proceeds to the next round once the dynamic parameters have been updated.

2.1.3 The Agents

In the valuation of DAO tokens, consideration of both trading benefits and utility values is vital. The agents in the DAO token market must account for the expected return from selling the tokens in the future as well as the value of the goods and services accessible through them. The agent problem proposed in Cong et al. (2021) is aligned with this objective, making it a suitable foundation for our approach. Therefore, the agents created in the proposed model behave based on the agent payoff function introduced in Cong et al. (2021), and we call them CLW agents. In our model, as in Cong et al. (2021), the agent rely on constant heterogeneous parameters. The total payoff function for a tokenized economy introduced by Cong et al. (2021) serves as the valuation mechanism for each agent i follows:

$$V(k) = dv_{i,t} = (P_t k_{t,i})^{1-a} (N_t A_t \exp(u_i))^a dt - \phi dt - P_t k_{t,i} r dt + P_t k_{t,i} \mu_{t,i}^p dt \quad (3)$$

where P_t denotes the price of the token at time t , $k_{t,i}$ denotes the amount of token that agent i is holding at time t , a is a constant defining diminishing utility, N_t is the number of agents who hold positive amount of tokens, ϕ denotes a participation cost, μ^p denotes a drift in price, and r denotes a risk-free rate. u_i is a heterogeneous parameter defining the perceived level of the DAO token's utility value which follows a normal distribution with controllable standard deviation (see Cong et al. (2021) for the argument about the distribution of u_i). Unlike the original specification, μ^p has the subscript i and t because each agent consider heterogenous window w_i to compute the expected return at time t using past price records. The window length w_i follows a uniform distribution with controllable parameters. This heterogeneous window length was adopted in Raberto et al. (2001) and Cocco et al. (2017) and captures agent's varying expectation on the market while introducing further complexity in the simulation.

In past studies of artificial financial markets, the fundamentalist-chartist design is popular and well-established (see Farmer & Joshi (2002), Jacob Leal et al. (2016), and Kirman (1993) for example). The fundamentalist-chartist approach is widely argued to reflect the macro-level behavior of financial markets by capturing the interplay between value-driven and trend-following strategies. This design assumes a heterogeneous balance among agents, where some focus on the fundamental, intrinsic value of tokens, while others prioritize trading benefits and speculative opportunities. The coexistence of these two behavioral archetypes fosters interactions that lead to complex endogenous dynamics, including price volatility, bubbles, and crashes. Our model is consistent with these studies, as the CLW agents may act as fundamentalist or chartist depending on their parameters. For a relatively large value of u_i , the fundamental value of the token as a means of economic activities within the platform is likely to be the dominant factor of that agent i 's token valuation. For a relatively small u_i , the expected risk premium is likely to be the dominant factor. By adopting the fundamentalist-chartist framework, we believe that the DAO token market is driven by fundamental values and speculative motivation, which is consistent with Rutskiy et al. (2023).

During each trading round, every agent decides to buy or sell the token. The agent first computes its optimal token holdings using the analytical solution proposed by Cong et al. (2021):

$$k^*_{t,i} = \frac{N_t A_t \exp(u_i)}{P_t} \left(\frac{1-a}{r - \mu^p_{t,i}} \right)^{\frac{1}{a}} \quad (4)$$

The agent then compares its current token holdings $k_{t,i}$, and decides on an action. If $k^*_{t,i} < k_{t,i}$, agent i places a buy order. If $k^*_{t,i} > k_{t,i}$, agent i places a sell order. If $k^*_{t,i} = k_{t,i}$, agent i remains inactive. The number of token bought or sold by each agent is restricted to c where c is an adjustable parameter (see Feng et al. (2021) for an example of similar design). Therefore, the agents aim to gradually adapt their token holdings to their optimal token holdings given the current market conditions. This normalization helps the model to avoid disproportionate impacts from certain agents in price formation, since the agents are not constrained by wealth parameters in our model, following Cong et al. (2021). The price at which the agent buys or sells is calculated based on equation (3). If $k^*_{t,i} < k_{t,i}$, the agent i places a buy order at the price $V(k_{t,i} + c) - V(k_{t,i})$. If $k^*_{t,i} > k_{t,i}$, the agent i places a sell order at the price $V(k_{t,i}) - V(k_{t,i} - c)$. The agents place these buy or sell orders based on the expected increase in payoff after trading. This represents willingness to pay for agents with $k^*_{t,i} < k_{t,i}$ and willingness to sell for agents with $k^*_{t,i} > k_{t,i}$. These orders are executed by the order book mechanism, as explained in the previous section. The price calculated by the order book will always be greater than or equal to 0.

Other than placing orders, agents can also engage in staking activities. Staking is a common practice in decentralized platforms, allowing the token holders to earn rewards by locking up the liquidity (Cong et al., 2022b). Staking can be interpreted as a sign of long-

term contribution and attachment to the platform, reflecting optimism in its future prospects. By locking up tokens, the tradable supply is reduced, which may help stabilize the token price through lower immediate sell pressure. In essence, agents who stake signal that they do not anticipate liquidating their holdings anytime near future, thereby reinforcing overall market confidence. In our model, we implement staking by interpreting it broadly as a signal that an agent is becoming a long-term investor. Every time agents obtain additional tokens or when their tokens are unlocked, the agents stake those tokens if $k^*_{t,i} < k_{t,i}$. To simplify the staking mechanism, we allow the model observers to exogenously define the portion of locked tokens to unlock at the end of each round.

2.1.4 The Token Supply

Cong et al. (2021) assumed a constant token supply. However, in real-world cryptocurrency markets, issuers often disclose plans regarding the total amount of tokens to be released and the mechanisms for their distribution in a whitepaper. Such schedules can significantly influence short-term token valuation and shape broader market dynamics. Therefore, we extend Cong et al. (2021) by introducing a dynamic token supply. During the simulation, tokens are supplied according to a predetermined schedule. The parameters of this token supply schedule are controllable to create an S-shaped supply schedule. The inflection point of the token supply schedule curve, the number of tokens issued per round, and the number of rounds in which the token issue amount is doubled or halved are the controllable parameters. By setting the inflection point to 0 or the last simulated period, it is also possible to produce a concave up or down token supply schedule.

Regarding token supply methods, offering tokens through exchanges or distributing them as rewards is widely practiced in many real-world token economies. For example, BitTorrent published a whitepaper through Binance, specifying how much of its token would be on sale and how much will be supplied as rewards (Binance, 2019). In our model, tokens can be issued via exchanges or as rewards. Tokens issued via exchange may be purchased at the agent's willing price. Therefore, market sell orders are inserted into the set of sell orders placed by the agents. Rewards are distributed randomly to agents who hold a positive token balance as a platform adoption reward. To avoid concentration of tokens, reward tokens are distributed to those random eligible agents as uniformly as possible.

After incorporating dynamic token supply and reward system, we denote the expected token reward as k^{reward} , which modifies the agent's buy and sell decisions. If $k^*_{t,i} < k_{t,i} + k^{reward}$, agent i places a buy order. If $k^*_{t,i} > k_{t,i} + k^{reward}$, agent i places a sell order. If $k^*_{t,i} = k_{t,i} + k^{reward}$, agent i remains inactive. If tokens are supplied through rewards, agents who satisfy reward conditions will consider the expected amount of reward when making trading decisions. It is assumed that agents are informed about how much tokens will be issued and with what probability they will be rewarded, allowing them to form rational expectations on the amount of reward they might receive this round.

2.1.5 The Market Growth

Following Cocco et al. (2017), to extend the Cong et al. (2021) model, we incorporate a dynamic agent population. However, unlike Cong et al. (2021) which

modeled serviceable market size as a constant and Cocco et al. (2017) which modeled a steady inflow of agents, our model allows observers to create a predetermined S-shaped schedule for the number of agents in the artificial market. This S-shaped curve can flexibly capture realistic user adoption patterns: slow initial intake, rapid growth, and saturation, which were patterns discovered by Cong et al. (2021). Furthermore, by controlling parameters, the observer is able to schedule a variety of market growth curves including linear, concave, convex growth curves. The parameters that control the S-shaped schedule are identical to token supply schedule parameters explained in Section 2.1.4. This extension of the Cong et al. (2021) model creates a necessity to change the unit of N_t . In Cong et al. (2021), N_t was as a measure ranging between 0 and 1, but to preserve the linear effect of N_t on the term $N_t A_t \exp(u_i)$ in equation (3) and (4), the range of N_t is modified to $[0, \infty)$ and it now represents the number of agents with $k_{t,i} > 0$.

2.2 Extending the Model: Incorporation of Governance Value

Cong et al. (2021) did not explicitly incorporate governance value in its token specification. An essential functionality of a DAO token is to empower token holders to collectively govern the platform. Since our primary focus is on a DAO token rather than just a platform native cryptocurrency, we extend Cong et al. (2021)'s specification and analytical solution. Specifically, we update equation (3) and derive a new analytical solution for optimal token holdings to replace equation (4). The derivation steps of the new optimal token holdings equation mirror those in Cong et al. (2021).

We first modify the total payoff function as follows:

$$V(k) = dv_{i,t} = (P_t k_{t,i})^{1-a} (N_t A_t \exp(u_i))^a dt - \phi dt - P_t k_{t,i} r dt + P_t k_{t,i} \mu_{t,i}^p dt + \frac{p^G k_{t,i} G_t dt}{M_t} \quad (5)$$

The only modification is the addition of the term $p^G k_{t,i} G_t dt / M_t$, where G_t is the total governance value of the circulating token supply at time t , $p_{t,i}^G$ is a binary variable representing agent i 's governance participation at time t , and M_t denotes the circulating token supply at time t . The term $p_{t,i}^G k_{t,i} G_t dt / M_t$ can be interpreted as the governance value of the tokens held by agent i . For simplicity, this specification assumes a linear voting mechanism. Using a derivation steps in Cong et al. (2021), the solution for the optimal token holdings becomes:

$$k_{t,i}^* = \frac{N_t A_t \exp(u_i)}{P_t} \left(\frac{1-a}{r - \mu_{t,i}^p - \frac{p_{t,i}^G G_t}{P_t M_t}} \right)^{\frac{1}{a}} \quad (6)$$

The new optimal token holdings equation now includes the term $p_{t,i}^G G_t / P_t M_t$, a governance value per unit of market capitalization, which reduces the carry cost of the tokens.

Following Johnson et al. (2019) and Xu et al. (2022), we compute the governance value of the tokens using the Discounted Cash Flow (DCF) approach:

$$G_t = CASH_t + \frac{TERMINAL}{(1+d)^T} \quad (7)$$

$$TERMINAL = \frac{CASH_t}{d} \quad (8)$$

where d is the discount rate, $CASH_t$ is the cash flow collected by the platform at time t , and $TERMINAL$ is the terminal value computed using the perpetuity method. The magnitude and growth of $CASH_t$ can be controlled by the observer.

In addition to including the governance value of tokens, our model also considers the voting participation rates. Low participation in DAO governance has frequently been noted, and it cannot be neglected when assessing governance value in DAO token valuation. An example of estimation for DAO voting participation can be found in Ding et al. (2023), which estimated the governance participation rate of fewer than 10%. We denote voting participation by $p_{t,i}^G$, where the probability $P(p_{t,i}^G = 1)$ is uniform among all agents in the artificial market. At the beginning of each trading round, agents decide whether to participate in the governance process that will be right after this trading round. If the agent decides to participate in governance, then the value of G_t computed by equations (7) and (8) applies to the valuation equation (5). On the other hand, if the agent decides not to participate in governance, then G_t will be set to 0 (which leads to equation (3)), meaning the agent does not care about the governance value of tokens during this round. For simplicity, this decision process assumes that the agent does not consider the possibility of participating in governance in the future. The model assumes that all agents participate in the governance process with equal probability.

2.3 Summary of Model Components

Now that all components of the model have been presented, Figure 3 summarizes the overall flow of the simulation and Figure 4 classifies each component in terms of exogeneity and endogeneity and illustrates their relations. In Figure 4, the token supply parameters introduced in Section 2.1.4, the market growth parameters introduced in Section 2.1.5, and agent payoff parameters introduced in Sections 2.1.1, 2.1.3, and 2.2 are shown as exogenous variables. The agent payoff parameters include the productivity measure A_t , a Cobb-Douglas constant a , the perceived utility value level constant u_i , the platform participation cost ϕ , the risk-free rate r , the heterogeneous window length w_i , voting participation $p_{t,i}^G$, and the discount rate d . These inputs are processed through the agent payoff function (equation (5)) and market mechanisms illustrated in Sections 2.1.2 and 2.1.3. The computation generates the DAO token value P_t and user adoption N_t , which

will be fed back into the agent payoff function in the next simulation period. In this way, observer's inputs indirectly affect the simulated DAO token value and platform user base, which grow endogenously as the simulation proceeds.

3. Calibration

3.1 Calibration Problem

The calibration of the proposed ABM is conducted to demonstrate the model's ability to align with empirical data, thereby improving its predictive accuracy and reliability. We formulate the calibration problem as follows:

$$\min_{\theta \in \Theta} f(\theta) \quad (9)$$

where θ is a vector of parameters to be calibrated, and Θ defines the space of parameter combinations to be explored. The objective function f is defined by equations (10) and (11):

$$\hat{G}(\theta) = \frac{1}{R} \sum_{s=1}^R [(m|\theta) - m] \quad (10)$$

$$f(\theta) = \hat{G}(\theta)^T W \hat{G}(\theta) \quad (11)$$

The objective function f represents the deviation of moments and statistics between real and simulated data. m is a vector of moments and statistics from the real data, $(m|\theta)$ is the simulated moments and statistics using the parameters θ , R is the number of simulation conducted using parameters θ , and W is the weight matrix for moments and statistics used as a criteria. The weight matrix W is computed by taking the inverse of the covariance matrix for moments and statistics from the bootstrapped data. For the choice of quadratic loss function and the weight matrix W , we mainly follow Platt & Gebbie (2016) which uses similar techniques to Winker et al. (2007), Fabretti (2013), Recchioni et al. (2015), and Zegadlo (2021).

The calibration criteria consist of the mean squared error (MSE) of price series, standard deviation, kurtosis, Hurst exponent. All criteria except for price MSE are adapted following Platt & Gebbie (2016), Fabretti (2013) and Platt (2020) so that the simulated data share similar characteristics with the real data. We include price deviation since the proposed model is a valuation model, and it must give an accurate forecast of the token value (see Recchioni et al. (2015), Fraunholz et al. (2021), Ghosh & Chinthalapati (2014) for example). MSE is a popular measure used in Recchioni et al. (2015) and ABM economic forecasting models (such as Young et al. (2014), Poledna et al. (2023), and Tubbenhauer et al. (2021).

Not all parameters of the proposed model are subject to calibration. Most variables can be inferred from the real-world data or assumed based on the observer's experience and knowledge. Only the essential parameters which are difficult to infer from the data are estimated. Specifically, we calibrate four parameters that significantly affect the model's

behavior: initial platform productivity A_0 , platform productivity drift μ^A , platform productivity diffusion σ^A , and the standard deviation of token utility level u^i .

Finally, the algorithm for calibration is similar to that of Gilli & Winker (2003) and Zegadlo (2021). A number of random parameter combinations are initially sampled from the considered range. Multiple simulations are performed for each parameter set, resulting data samples that are then evaluated and averaged to assess performance. We filter to retain a portion of well-performing ones. Only the neighbors of the selected ones are used in the next round. This procedure is repeated until all computational budget are consumed or until no better parameter set is found. Considering the available budget, each parameter set is simulated 30 times, and the top 10~5% of the well-performing parameter sets are considered for the next step.

3.2 A Calibration Case Study: Ethereum ETH token

Ethereum ETH token serves as the case study. For this calibration, we use five years of early ETH data. The real ETH price, token holder, and circulating supply data spanning from 2015/8/7 to 2020/8/6 are extracted from Messari (n.d.). Table 1 summarizes the values assumed for each parameter and the range of values searched in the calibration process for ETH.

Parameter	Value/Range
Initial Platform Productivity (in log10)	[-10, 4]
Platform Productivity Drift	[0, 4]
Platform Productivity Diffusion	[0, 10]
Standard Deviation of u^i	[0, 10]
Initial Token Price	3
Initial Market Size	100
Inflection Period for Market Growth Curve	116.579825
Market Growth Rate (Before Inflection)	8.31137145
Market Growth Rate Doubling Period (Before Inflection)	992.567141
Market Growth Rate (After Inflection)	60.3736513
Market Growth Rate halving Rounds (After Inflection)	9999.94059

Initial Token Supply	5201438
Inflection Period for Token Supply Schedule	239.20867425
Token Issue Rate (Before Inflection)	6234.13173191
Token Issue Rate Doubling Period (Before Inflection)	54592.74306914
Token Issue Rate (After Inflection)	5327.49443937
Token Issue Rate halving Rounds (After Inflection)	324.0944231
Annual Platform Revenue per User	23.67
Ratio of Tokens Supplied via Reward	0.5
Ratio of Tokens Supplied via Exchange	0.5
Governance Participation Rate	0.1
Discount rate	0.3
Window length	7 ~ 365 days uniformly distributed among agents
Risk-free rate	0.05
Intra-period (month) trading rounds	10

Table 1. The value and range of parameters to be considered in the ETH case study.

The calibration parameters are set across a broad range to encompass various potential scenarios, ensuring multiple model runs produce simulated data closely aligned with real-world observations. The initial price is set to the Ethereum price recorded on August 7, 2015. Token allocation ratios for different supply methods are configured to approximately match those proposed by Buterin (2014) and the tokenomics information available in Messari (n.d.). The annual discount rate, risk-free rate, intra-period trading rounds and window length distribution parameters are configured as shown in Table 1. Since the simulation period is measured in monthly intervals, 10 trading rounds per period indicates that agents are allowed to trade their tokens up to 10 times per month. The window length ranges from 1 week to 1 year, assigned randomly to agents to incorporate heterogeneous expectation. Lastly, a constant α which defines a diminishing utility from token holdings is fixed at 0.25, close to the value originally used in Cong et al. (2021). These parameter assumptions aim to conserve computational resources and make the calibration problem less complex.

The market growth parameters and token supply schedule parameters are configured to approximate historical ETH data. The cumulative market size curve and token supply schedule are assumed to be linear or concave down mirroring the trends observed in historical data on ETH holders and supply. The initial market size is fixed at 100, as simulation using actual token holder counts requires a substantial level of computational resources. Cocco et al. (2017) took the similar approach, scaling down the simulated market by 100. Given this initial market size of 100, the market growth rate parameter is chosen so that the simulated ratio between the initial market size and market size upon termination matches empirical data. Figure 5 compares the simulated and actual market growth. The token supply schedule parameters are then selected to align with historical data on the ratio of token holders to circulating supply. The comparison of simulated versus actual token supply, as well as holders-to-supply-ratio, is depicted in Figures 6 and 7, respectively. The objective of this parameter configuration approach is to preserve the realistic demand-supply dynamics while reducing overall computational requirements for calibration.

3.3 Results

The parameter set chosen as a result of calibration is illustrated in the Table2.

Parameter	Estimated value
Initial Platform Productivity	-0.710
Platform Productivity Drift	2.619
Platform Productivity Diffusion	2.589
Standard Deviation of Token Utility Parameter	2.512

Table 2. Estimated value of parameters.

Table 2 summarizes the calibration results after 42,000 simulations in total. The initial value, drift, and diffusion parameters for the geometric Brownian process of platform productivity were estimated as -0.710, 2.619, and 2.589 respectively. Figure 8 compares the actual ETH price series with the central price forecast from the model along with its 15% percentile range. Simulations generated by the calibrated model capture the price levels except for precise price spike timing. Although the model predicts that price spikes will occur during the simulation, the timing of these spikes remains too uncertain to forecast. The calibrated model's ability to capture the overall trend and potential price spikes provides the observer with an estimate of the token's value while accounting for uncertainty, which is essential for valuation purposes.

4. Conclusion

4.1 Novelty and Application

This research proposes an agent-based framework in the field of cryptocurrency valuation, specifically for DAO tokens with both governance and utility values. Using the stochastic model proposed in Cong et al. (2021) as a foundation, our model incorporates dynamic token supply, dynamic market size, token locking, and heterogeneous expectations. While valuation of DAOs in an early stage of an IEO has seldom been explored particularly together with various tokenomics designs, our model addresses these overlooked aspects. In addition, the simultaneous consideration of trading benefits, utility values, and governance value is rarely seen in existing literature. The parameters of our proposed model allow for consideration of various tokenomics designs, a topic which has been rarely studied. A price formation mechanism driven by artificial financial agents interacting and a simulated order book captures the nuanced economic behavior and complexity of the cryptocurrency market unlike traditional equation-based approaches. By doing so, this framework can potentially offer a practical way to gauge a fair valuation for DAO tokens awaiting IEO, opening up new possibilities for research and industry applications. The advantages of equation-based approaches are notable, but we aim to open and stimulate new directions to study the cryptocurrency market and DAO tokenomics.

4.2 Limitations and Extensions

The research presented is not without limitations. Incorporating complexity comes with a cost of scalability issues and computational intensity. Obtaining forecasts and calibration results requires a substantial amount of computational power compared with the traditional equation-based approaches. This limitation can be alleviated by training a machine learning surrogate of the proposed model (as in Zhang et al. (2020)), which can be a future extension to this paper. One approach is to use a generative model to capture the distribution of the simulation results, since our agent-based model relies on the stochastic processes to incorporate the volatile nature of the cryptocurrency markets as in Cong et al. (2021). It is worth noting that although a surrogate approach can reduce scalability and computational issues, it may fail to capture the full emergent structures arising from the an agent-based complex system.

Furthermore, the model offers a promising framework for optimizing token supply parameters to help prevent sharp price drops during the early stages of a DAO token launch. By adjusting the token supply curve and the associated supply method parameters within the simulation, stakeholders can systematically evaluate how different tokenomics designs impact token value trajectories over time. In practice, the model can generate synthetic data to investigate these effects, enabling researchers and practitioners to explore a wide parameter space and identify configurations that promote price stability, encourage user adoption, or meet other specific objectives. This approach is particularly significant given that the design of DAO tokenomics often demands interdisciplinary expertise—from economics and game theory to political science, management, control theory, and psychology—a combination that can be challenging to assemble. Consequently, this agent-based framework stands out as a powerful statistical tool for DAO platformers, supporting evidence-based decision-making in the complex realm of tokenomics design.

One possible extension of the presented agent-based framework is further customization of the agent payoff function to better accommodate platform-specific features of different DAOs. The current payoff function and the derived optimal token holding formula are intentionally broad and were generalized based on Cong et al. (2021). While this formulation effectively captures a wide range of market dynamics, its application is heavily reliant on calibration results that often borrow parameters from already launched platforms, implicitly assuming that the utility level offered by the pre-launch platform is similar to that of the reference platform. For industry applications, it would be beneficial to tailor the payoff function with assumptions that more accurately reflect the unique characteristics and utility levels of the targeted DAO. For example, one can incorporate Non-Fungible Token (NFT) based incentivization mechanisms such as the one explored in Fukuhara & Sakai (*unpublished work*) in the simulation as one of the DAO-specific features. In this context, the integration of generative AI could be particularly valuable; it would allow DAO platformers, including those without extensive coding expertise, to simulate token valuation with more specialized agent specifications. Importantly, even if the payoff function is adapted to a more specialized version, the model's overall framework and underlying logic explained in Section 2.3 remain preserved, ensuring its robustness and broad applicability across various tokenomics designs.

Future research will investigate the endogenous staking mechanisms, different price formation mechanisms, the incorporation of user networks, diverse voting mechanisms, income distributions, and dependencies among the cryptocurrency market. Enriching and extending the calibration case studies to a wider variety of DAOs is also crucial for further validation and for the application of the model in initial token valuation.

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Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work the author(s) used ChatGPT in order to enhance the readability of the work. After using this tool/service, the author(s) reviewed and edited the content as needed and take(s) full responsibility for the content of the published article.

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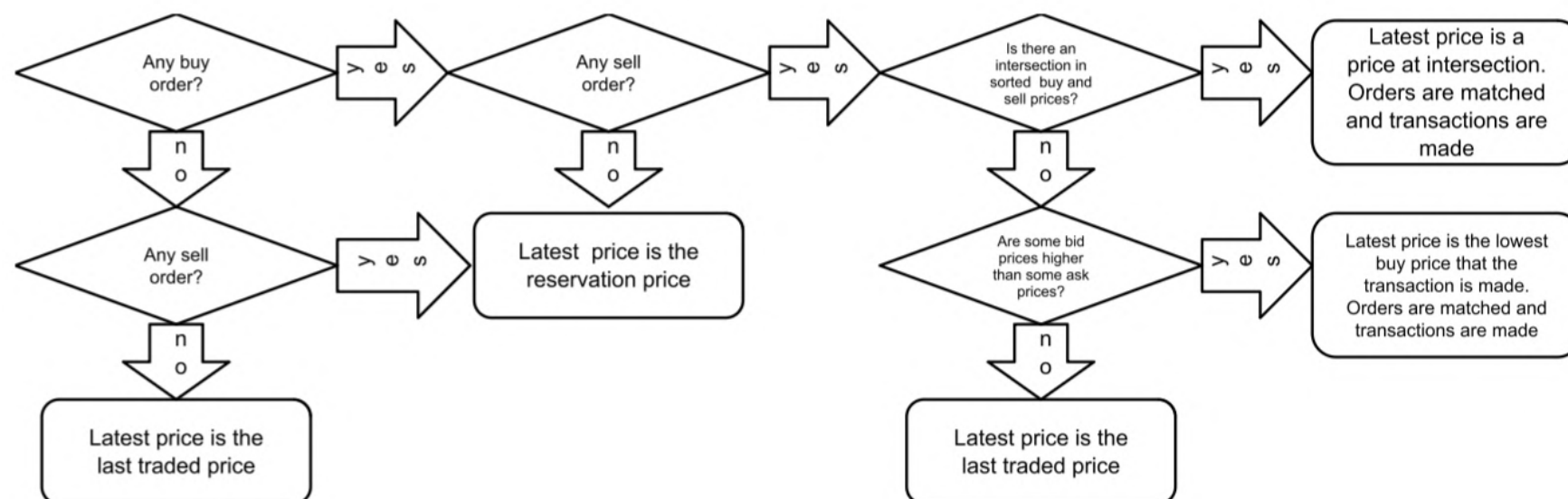


Figure 1. Calculation of the latest price using simplified order book.

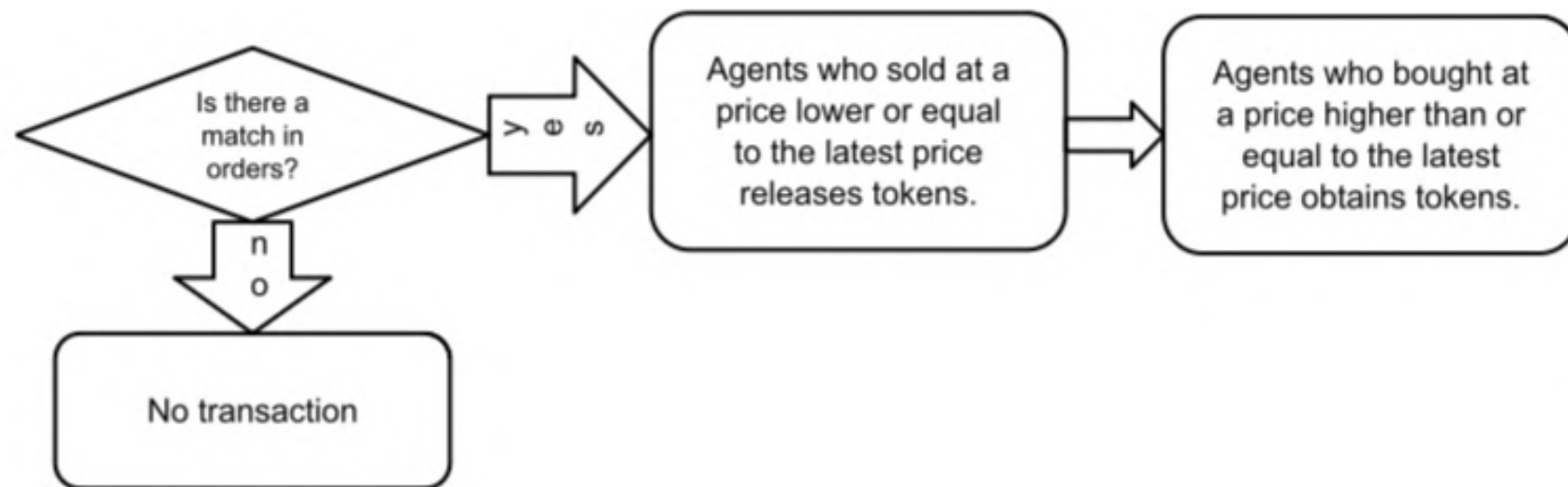


Figure 2. Execution of the orders based on the calculated latest price.

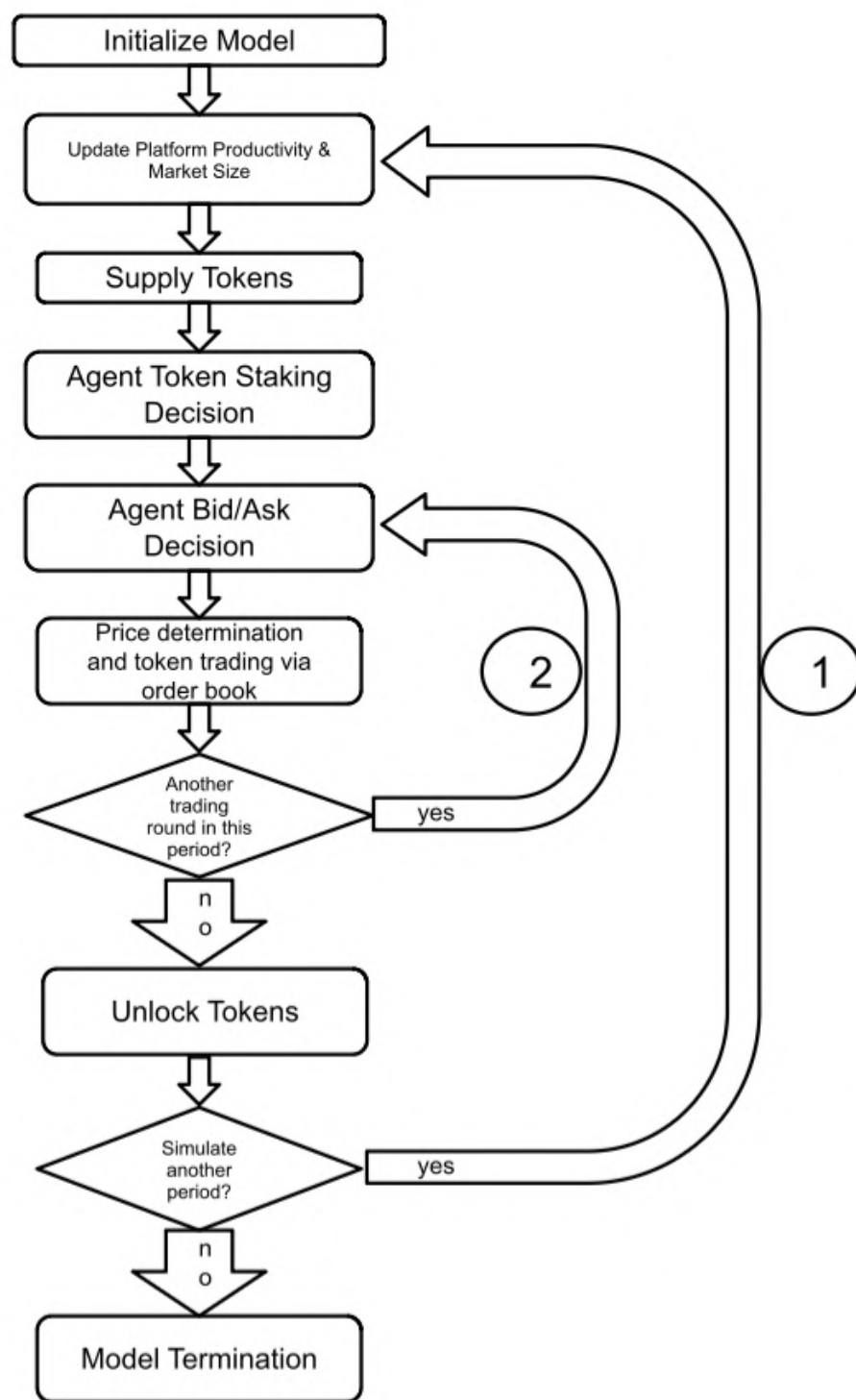


Figure 3. The overall flow of the model. The number of loop ① corresponds to how many trading rounds are simulated within a unit of period, and the number of loop ② corresponds to how many periods are being simulated.

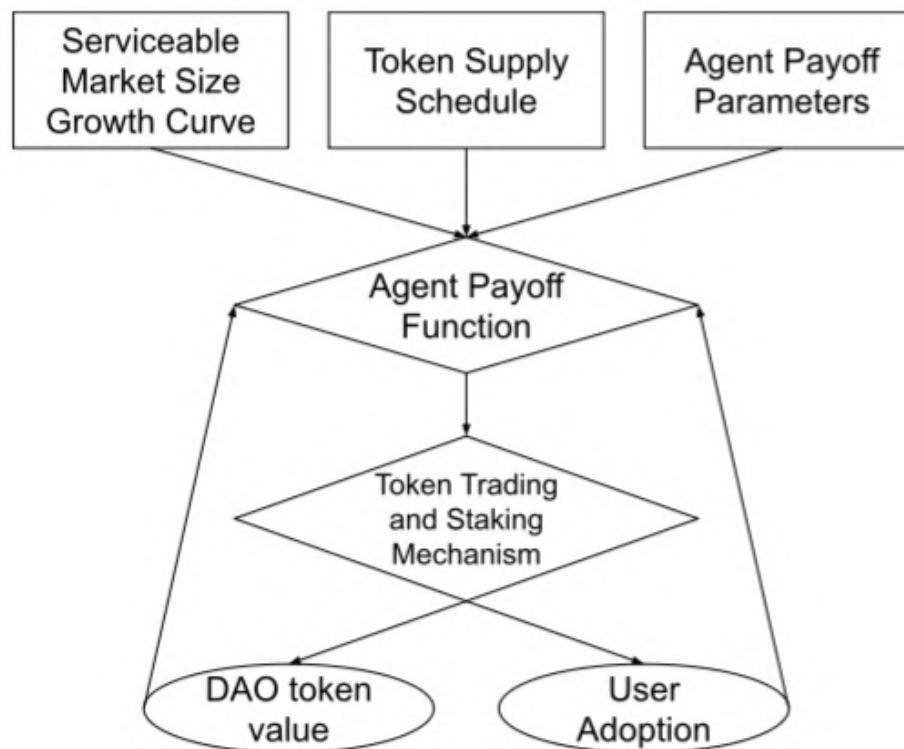


Figure 4. A relation map of the variables in the model. The variables shown as circles are endogenously determined during the simulation, while the variables shown as squares are determined exogenously as assumptions by the model observer. The components shown as diamond shapes are the set of computations. The arrows represent the dependencies among the components; the component from which an arrow originates directly affects the computation of the component to which it points.

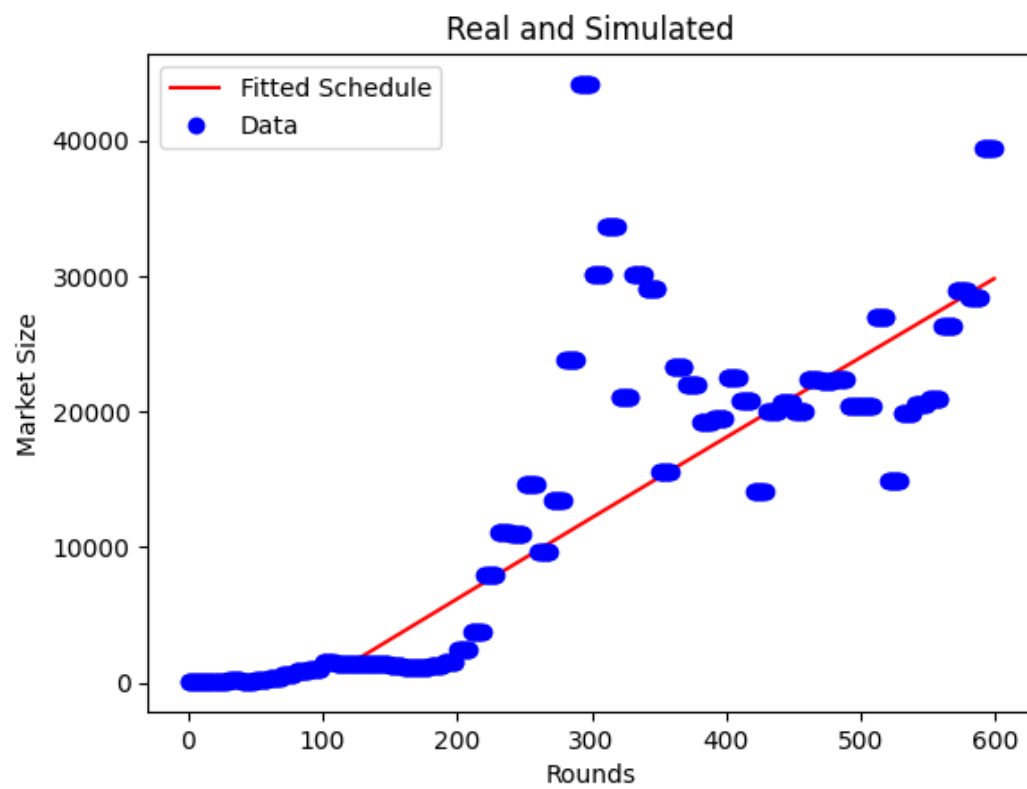


Figure 5. Comparison of simulated and actual market size (unique addresses count) growth.

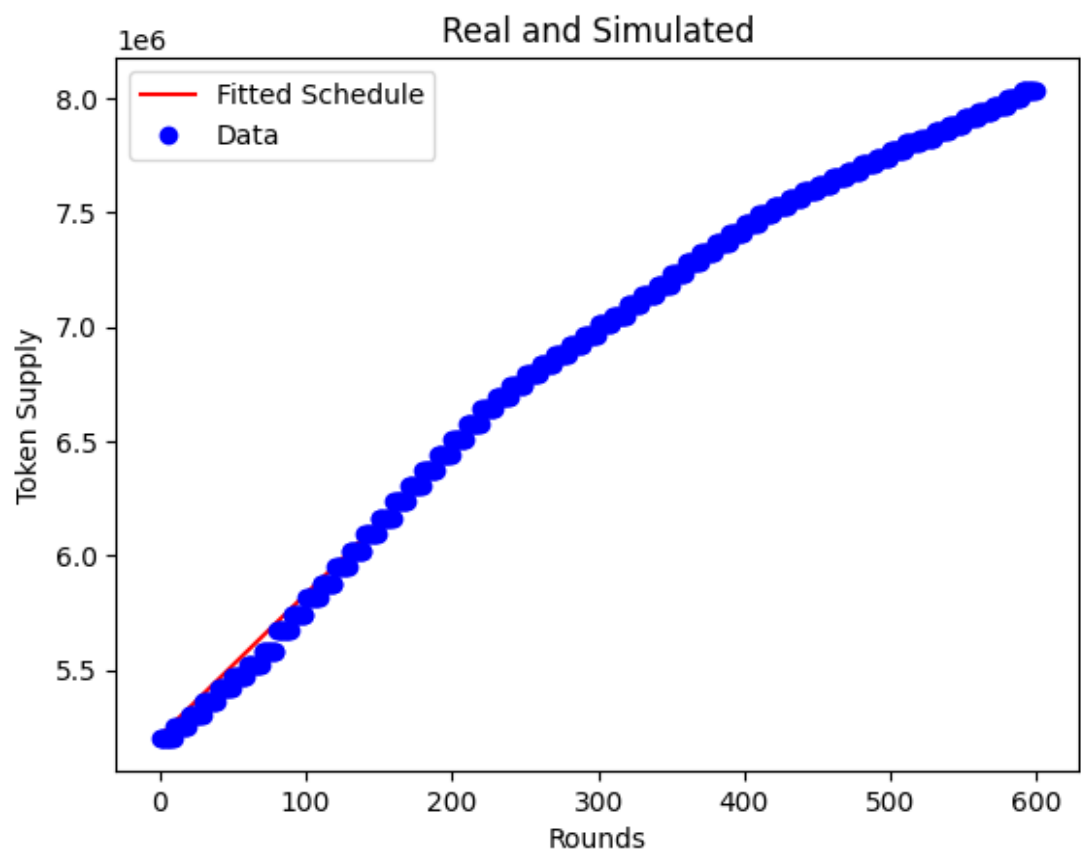


Figure 6. Comparison of simulated and actual circulating token supply.

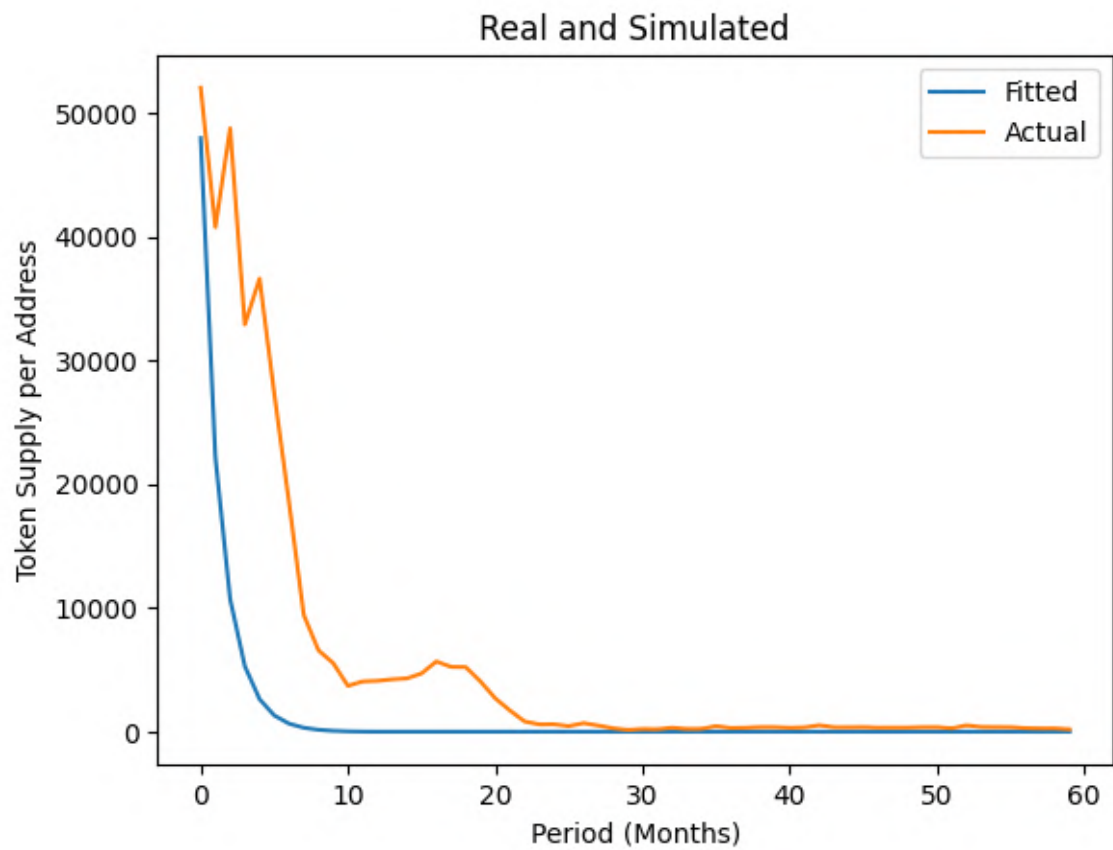


Figure 7. Comparison of simulated and actual token supply-holder (market size) ratio.

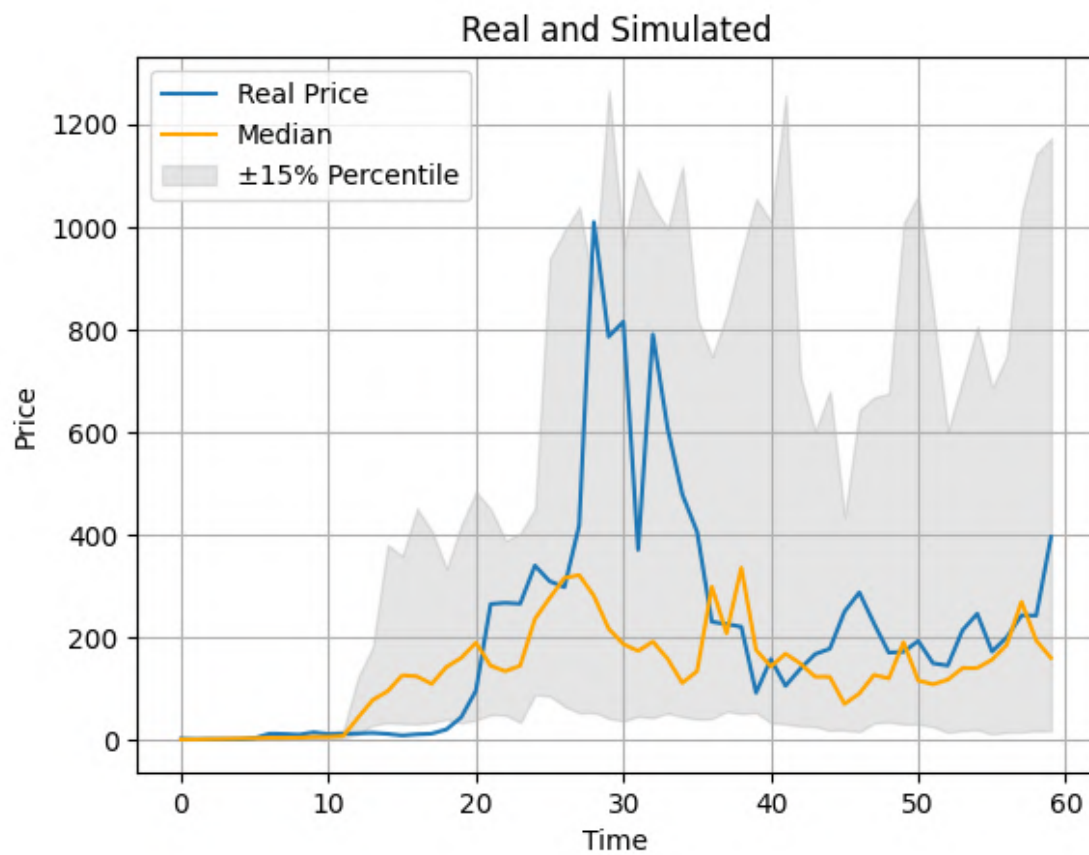


Figure 8. Comparison of actual ETH price series and the central (median) prediction of ETH price series by the calibrated model. The gray area represents a $\pm 15\%$ percentile range from the central prediction.